

CSEC Mathematics
June 2017 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the EXACT value of

(i) $\left(4\frac{1}{3} - 1\frac{2}{5}\right) \div \frac{4}{15}$ [2]

$$\begin{aligned} \left(4\frac{1}{3} - 1\frac{2}{5}\right) \div \frac{4}{15} &= \left(\frac{13}{3} - \frac{7}{5}\right) \div \frac{4}{15} \\ &= \left(\frac{(5 \times 13) - (7 \times 3)}{15}\right) \div \frac{4}{15} \\ &= \left(\frac{65 - 21}{15}\right) \div \frac{4}{15} \\ &= \frac{44}{15} \div \frac{4}{15} \\ &= \frac{44}{15} \times \frac{15}{4} \\ &= 11 \end{aligned}$$

(ii) $\frac{(3.1 - 1.15)^2}{0.005}$ [2]

$$\begin{aligned} \frac{(3.1 - 1.15)^2}{0.005} &= \frac{(1.95)^2}{0.005} \\ &= \frac{3.8025}{0.005} \\ &= 760.5 \end{aligned}$$

(b) A store is promoting a new mobile phone under two plans: Plan A and Plan B.

The plans are advertised as shown in the table below.

	Plan A	Plan B
Deposit	\$400	\$600
Monthly instalment	\$65	\$80
Number of months to repay	12	6
Tax on ALL payments	0%	5%

(i) Calculate the TOTAL cost of a phone under Plan A. [2]

Total cost of phone

= Deposit + (Monthly instalments $\times$ number of months) + Tax

= \$400 + (\$65 $\times$ 12) + \$0

= \$400 + \$780

= \$1180

$\therefore$ The total cost of a phone under Plan A is \$1180.

(ii) Determine which of the two plans, A or B, is the better deal.

Justify your answer.

[2]

Total cost of phone

= Deposit + (Monthly instalments $\times$ number of months) + Tax

$$= \$600 + (\$80 \times 6) + 0.05(600 + (80 \times 6))$$

$$= \$600 + \$480 + 0.05(600 + 480)$$

$$= \$1134$$

$\therefore$ Plan B is the better deal as it has an overall cost

$(\$1180 - \$1134 = \$46)$ cheaper than that of Plan A.

(c) John's monthly electricity bill is based on the number of kWh of electricity that he consumes for that month. He is charged \$5.10 per kWh of electricity consumed. For the month of March 2016, two meter readings are displayed in the table below.

	Meter Readings (kWh)
Beginning 01 March	0 3 0 1 1
Ending 31 March	0 3 3 0 7

- (i) Calculate the TOTAL amount that John pays for electricity consumption for the month of March 2016. [2]

Number of kWh used

= Final reading on 31st March – Initial reading on 1st March

$$= 0\ 3\ 3\ 0\ 7 - 0\ 3\ 0\ 1\ 1$$

$$= 296\text{ kWh}$$

If 1 kWh costs = \$5.10,

Then, 296 kWh cost = $\$5.10 \times 296$

$$= \$1509.60$$

$\therefore$ John pays \$1509.60 for electricity consumption for the month of March 2016.

- (ii) For the next month, April 2016, John pays \$2351.10 for electricity consumption.

Determine his meter reading at the end of April 2016. [2]

$$\begin{aligned} \text{Number of kWh used on April} &= \frac{\$2351.10}{\$5.10} \\ &= 461 \text{ kWh} \end{aligned}$$

$$\begin{aligned} \text{At the end of April, the meter reading should read} &= 03307 + 461 \\ &= 03768 \end{aligned}$$

Total: 12 marks

2. (a) Factorize the following expressions completely.

(i) $6y^2 - 18xy$ [2]

$$6y^2 - 18xy = 6y(y - 3x)$$

(ii) $4m^2 - 1$ [1]

$$4m^2 - 1$$

$$= (2m)^2 - (1)^2 \quad \text{[difference of two squares]}$$

$$= (2m + 1)(2m - 1)$$

(iii) $2t^2 - 3t - 2$ [2]

$$2t^2 - 3t - 2$$

$$= 2t^2 - 4t + t - 2$$

$$= 2t(t - 2) + 1(t - 2)$$

$$= (2t + 1)(t - 2)$$

(b) Write as a single fraction and simplify

$$\frac{5p+2}{3} - \frac{3p-1}{4} \quad [2]$$

$$\frac{5p+2}{3} - \frac{3p-1}{4}$$

$$= \frac{4(5p+2) - 3(3p-1)}{12}$$

$$= \frac{20p+8-9p+3}{12}$$

$$= \frac{20p-9p+8+3}{12}$$

$$= \frac{11p+11}{12}$$

$$= \frac{11(p+1)}{12}$$

(c) A formula is given as $d = \sqrt{\frac{4h}{5}}$.

- (i) Determine the value of d when $h = 29$. Give your answer correct to 3 significant figures. [2]

$$d = \sqrt{\frac{4h}{5}}$$

When $h = 29$,

$$d = \sqrt{\frac{4(29)}{5}}$$

$$= \sqrt{\frac{116}{5}}$$

$$d = \sqrt{23.2}$$

$$d = 4.82 \quad (\text{to 3 significant figures})$$

- (ii) Make h the subject of the formula.

$$d = \sqrt{\frac{4h}{5}}$$

Squaring both sides gives:

$$d^2 = \frac{4h}{5}$$

$$5d^2 = 4h$$

$$h = \frac{5d^2}{4}$$

Total: 11 marks

3. (a) The universal set U is defined as follows:

$$U = \{x: x \in \mathbb{N}, 2, x < 12\}.$$

The sets M and R are subsets of U such that

$$M = \{\text{odd numbers}\}$$

$$R = \{\text{square numbers}\}$$

- (i) List the members of the subset M . [1]

The universal set is $U = \{3, 4, 5, 6, 7, 8, 9, 10, 11\}$.

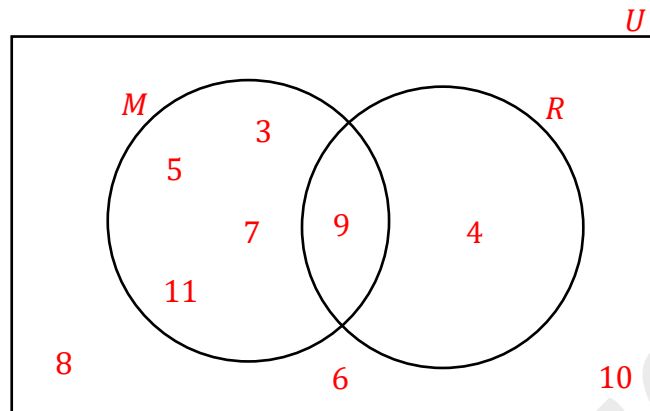
The members of set $M = \{\text{odd numbers}\}$
 $= \{3, 5, 7, 9, 11\}$

- (ii) List the members of the subset R . [1]

The members of set $R = \{\text{square numbers}\}$
 $= \{4, 9\}$

- (iii) Draw a Venn diagram that represents the relationship among the defined subsets of U . [4]

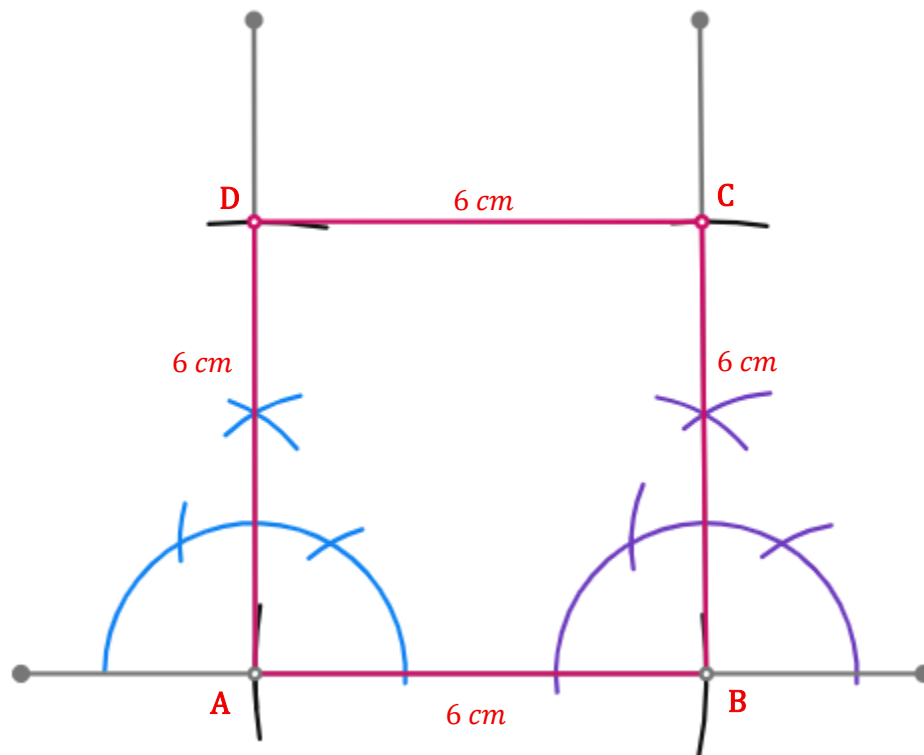
The Venn diagram is as follows:



(b) Using a ruler, a pencil and a pair of compasses,

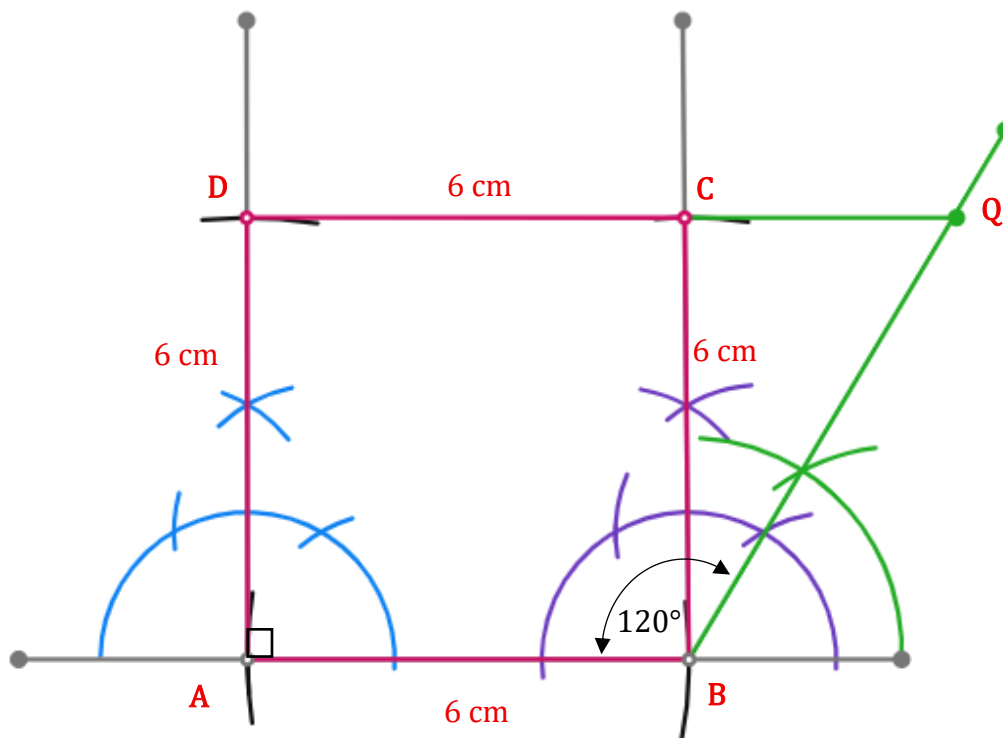
(i) construct accurately, the square $ABCD$, with sides 6 cm

[3]

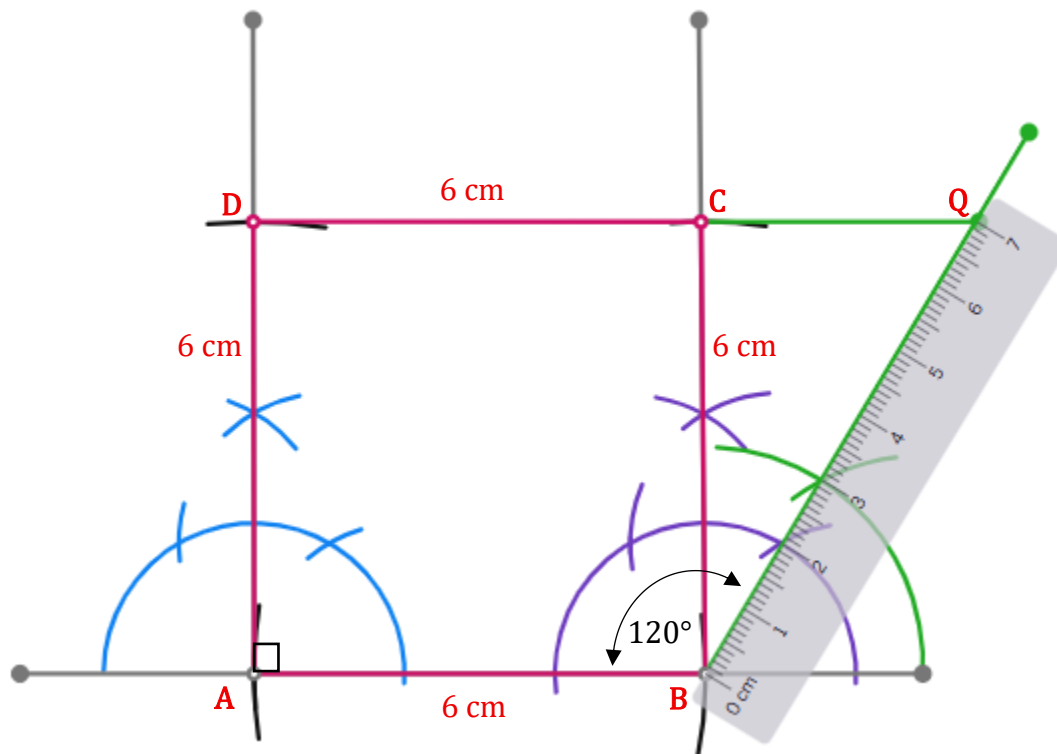


- (ii) construct, as an extension of your drawing in (b)(i), the trapezium $DABQ$ so that $\angle ABQ = 120^\circ$. [2]

[Note: Credit will be given for clearly drawn construction lines.]



(iii) Hence, measure and state the length of BQ .



The length of BQ is 7 cm .

Total: 12 marks

4. (a) The function f is defined as $f(x) = \frac{1}{3}x - 2$.

(i) Find the value of $f(3) + f(-3)$. [2]

$$f(3) = \frac{1}{3}(3) - 2$$

$$= 1 - 2$$

$$= -1$$

$$f(-3) = \frac{1}{3}(-3) - 2$$

$$= -1 - 2$$

$$= -3$$

$$\therefore f(3) + f(-3) = -1 + (-3)$$

$$= -1 - 3$$

$$= -4$$

(ii) Calculate the value of x for which $f(x) = 5$. [2]

$$f(x) = 5$$

So, we have,

$$\frac{1}{3}x - 2 = 5$$

$$\frac{1}{3}x = 5 + 2$$

$$\frac{1}{3}x = 7$$

$$x = 7 \times 3$$

$$x = 21$$

(iii) Determine the inverse function $f^{-1}(x)$. [2]

$$f(x) = \frac{1}{3}x - 5$$

Let $y = f(x)$.

$$y = \frac{1}{3}x - 5$$

Interchanging the variables x and y .

$$x = \frac{1}{3}y - 5$$

Making y the subject of the formula.

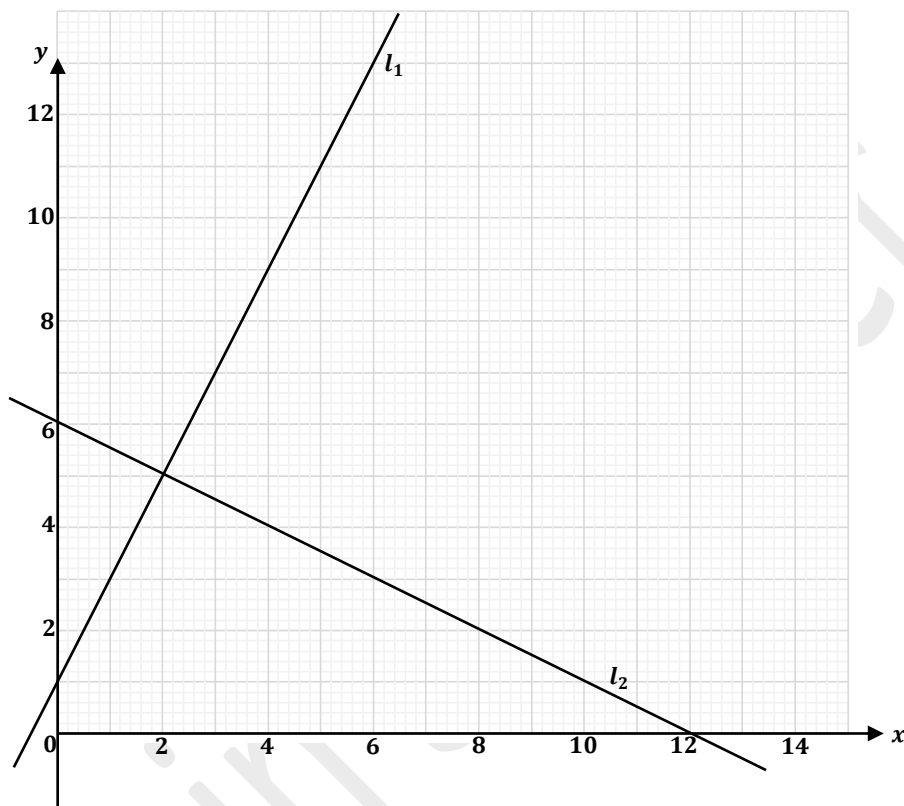
$$x + 5 = \frac{1}{3}y$$

$$y = 3(x + 5)$$

$$y = 3x + 15$$

$$\therefore f^{-1}(x) = 3x + 15$$

- (b) The graph below shows two straight lines, l_1 and l_2 . Line l_1 intercepts the y -axis at $(0, 1)$. Line l_2 intercepts the x and y axes at $(12, 0)$ and $(0, 6)$ respectively.



- (i) Calculate the gradient of the lines l_1 and l_2 . [2]

For the line l_1 : Points are $(0, 1)$ and $(2, 5)$.

$$\text{Gradient of line } l_1 = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - 1}{2 - 0}$$

$$= \frac{4}{2}$$

$$= 2$$

For the line l_2 : Points are (12, 0) and (0, 6).

$$\begin{aligned}\text{Gradient of line } l_2 &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{6 - 0}{0 - 12} \\ &= \frac{6}{-12} \\ &= -\frac{1}{2}\end{aligned}$$

- (ii) Determine the equation of the line l_1 . [2]

The general equation of a straight line is of the form $y = mx + c$.

Gradient of line $l_1 = 2$

y -intercept of line $l_1 = 1$

$\therefore$ The equation of the line l_1 is $y = 2x + 1$.

- (iii) What is the relationship between l_1 and l_2 ? Give a reason for your answer. [2]

Gradient of line $l_1 = 2$ and Gradient of line $l_2 = -\frac{1}{2}$

The gradient of line l_2 is the negative reciprocal of the gradient of line l_1 .

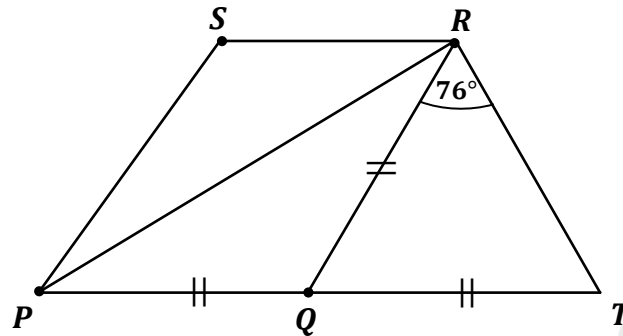
$\therefore$ Line l_1 is perpendicular to line l_2 .

Total: 12 marks

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5. (a) $PTRS$, **not drawn to scale**, is a quadrilateral. Q is a point on PT such that

$QT = QR = QP$ and $\text{Angle } QRT = 76^\circ$.



Determine, giving a reason for each step of your answer, the measure of

- (i) angle RQT [2]

The base angles of the isosceles triangle RQT are equal.

$$\therefore \text{Angle } RTQ = 76^\circ$$

The sum of the interior angles in a triangle add up to 180° .

$$\therefore \text{Angle } RQT = 180^\circ - (76^\circ + 76^\circ)$$

$$= 28^\circ$$

- (ii) angle PRT [2]

The base angles of the isosceles triangle RQP are equal.

$$\therefore \text{Angle } QPR = \text{Angle } QRP$$

The exterior angle of a triangle is equal to the sum of the interior opposite angles.

$$\text{Angle } QPR + \text{Angle } QRP = 28^\circ$$

$$\begin{aligned}\text{Angle } QRP &= \frac{28^\circ}{2} \\ &= 14^\circ\end{aligned}$$

Hence,

$$\begin{aligned}\text{Angle } PRT &= \text{Angle } QRP + \text{Angle } QRT \\ &= 14^\circ + 76^\circ \\ &= 90^\circ\end{aligned}$$

(iii) angle SPT , given that angle $SRT = 145^\circ$ and angle $PSR = 100^\circ$ [2]

$$\begin{aligned}\text{Angle } SRP &= 145^\circ - 90^\circ \\ &= 55^\circ\end{aligned}$$

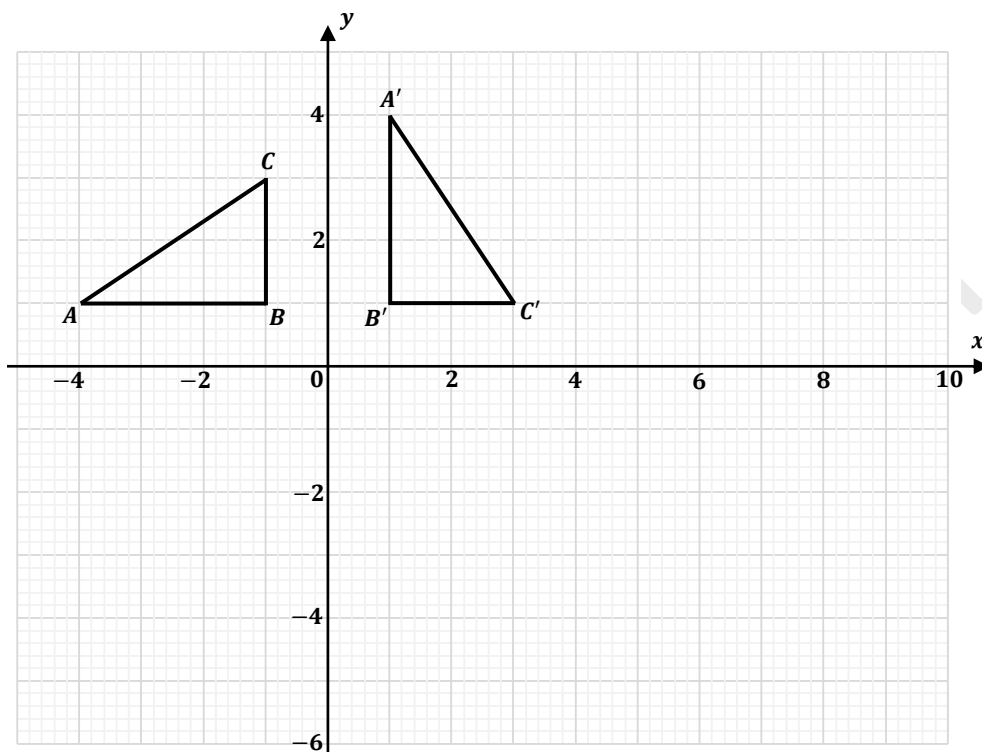
The sum of the interior angles in a triangle is 180° .

$$\begin{aligned}\text{Angle } SPR &= 180^\circ - (100^\circ + 55^\circ) \\ &= 25^\circ\end{aligned}$$

Hence,

$$\begin{aligned}\text{Angle } SPT &= 25^\circ + 14^\circ \\ &= 39^\circ\end{aligned}$$

- (b) The diagram below shows triangle ABC and its image, $A'B'C'$, under a single transformation.



- (i) Describe **completely** the transformation that maps $\triangle ABC$ to $\triangle A'B'C'$.

[3]

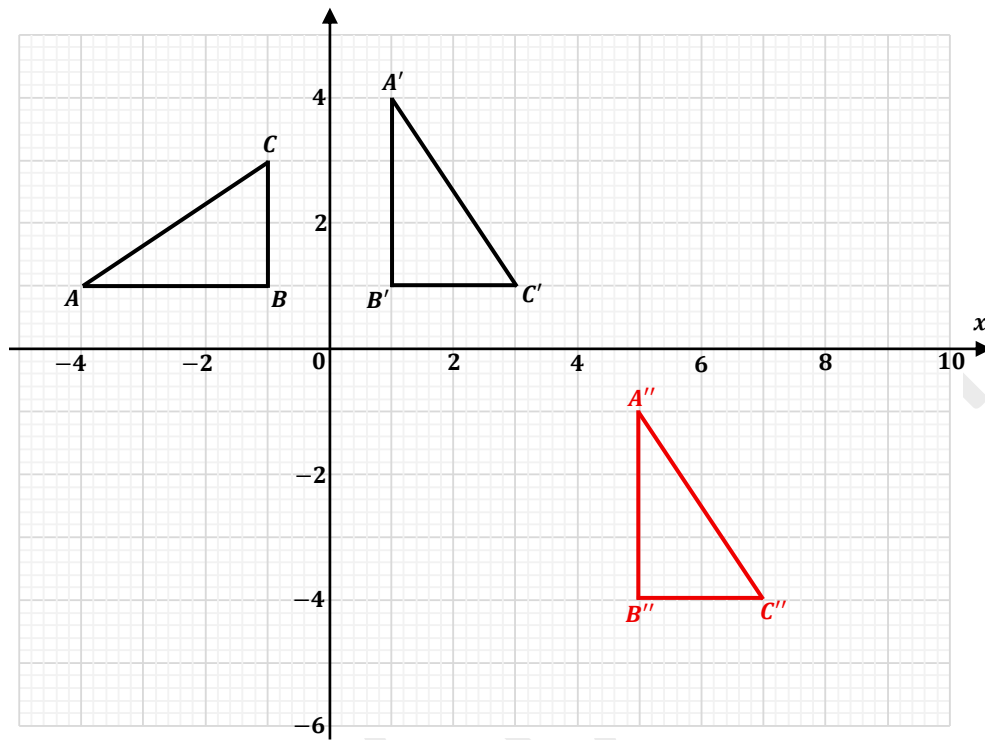
$A'B'C'$ is a 90° clockwise rotation of ABC about the origin, O .

- (ii) The translation vector $T = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$ maps $\triangle A'B'C'$ to $\triangle A''B''C''$.

On the diagram above, draw the $\triangle A''B''C''$.

[2]

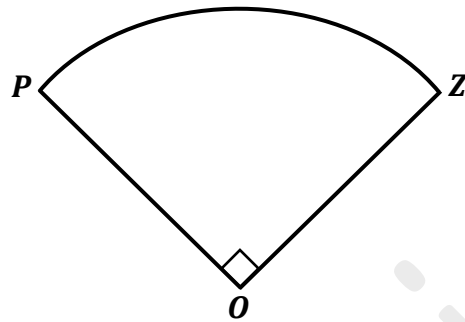
Shown on the diagram below:



Total: 11 marks

6. (a) In this problem, take π to be $\frac{22}{7}$.

The diagram below, **not drawn to scale**, shows a field in the shape of a sector of a circle with centre O and diameter 28 m . Angle POZ is 90° .



Calculate

- (i) the area of the field

[2]

$$\begin{aligned}\text{Radius} &= \frac{\text{Diameter}}{2} \\ &= \frac{28}{2} \\ &= 14\text{ m}\end{aligned}$$

$$\begin{aligned}\text{Area of the field} &= \frac{\theta}{360^\circ} \pi r^2 \\ &= \frac{90^\circ}{360^\circ} \times \frac{22}{7} \times (14)^2 \\ &= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 \\ &= 154\text{ m}^2\end{aligned}$$

$\therefore$ The area of the field is 154 m^2 .

(ii) the perimeter of the field

[3]

Perimeter of the field = Length of arc PZ + $2r$

$$= \left(\frac{\theta}{360^\circ} \times \pi d \right) + 2r$$

$$= \left(\frac{90^\circ}{360^\circ} \times \frac{22}{7} \times 28 \right) + 2(14)$$

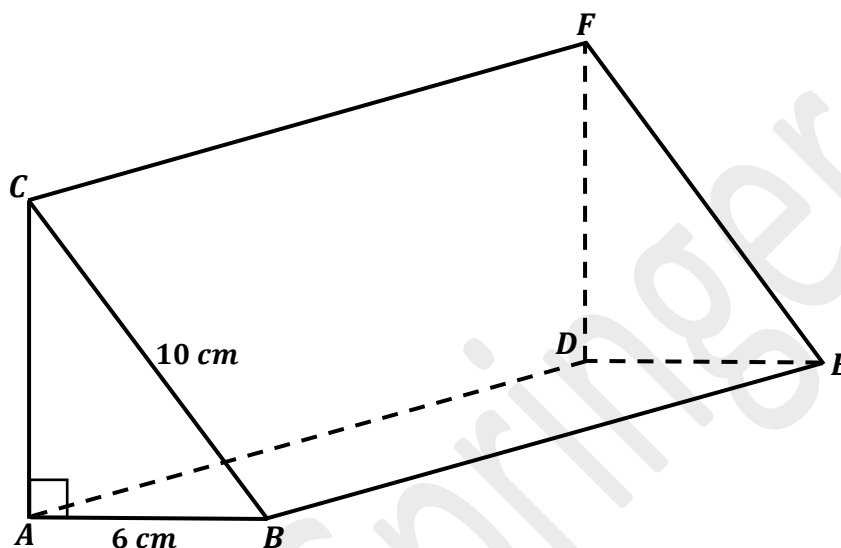
$$= 22 + 28$$

$$= 50 \text{ m}$$

$\therefore$ The perimeter of the field is 50 m .

(b) The diagram below, **not drawn to scale**, shows a triangular prism $ABCDEF$.

The cross section is the right-angled triangle, ABC , where $AB = 6$ cm and $BC = 10$ cm.



Calculate

- (i) the area of the triangle ABC

[2]

Using Pythagoras' Theorem,

$$BC^2 = AC^2 + AB^2$$

$$(10)^2 = AC^2 + (6)^2$$

$$100 = AC^2 + 36$$

$$AC^2 = 100 - 36$$

$$AC^2 = 64$$

$$AC = \sqrt{64}$$

$$AC = 8 \text{ cm}$$

Now,

$$\text{Area of } \triangle ABC = \frac{bh}{2}$$

$$= \frac{6 \times 8}{2}$$

$$= \frac{48}{2}$$

$$= 24 \text{ cm}^2$$

$\therefore$ The area of the triangle ABC is 24 cm^2 .

- (ii) the length of the prism, if the volume is 540 cm^3 [2]

Volume of the prism = Area of cross-section $\times$ Length of prism

$$540 = 24 \times \text{Length of prism}$$

$$\text{Length of prism} = \frac{540}{24}$$

$$\text{Length of prism} = 22.5 \text{ cm}$$

$\therefore$ The length of the prism is 22.5 cm .

(iii) the surface area of the prism

$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{bh}{2} \\ &= \frac{6 \times 8}{2} \\ &= \frac{48}{2} \\ &= 24 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } \triangle DEF &= \frac{bh}{2} \\ &= \frac{6 \times 8}{2} \\ &= \frac{48}{2} \\ &= 24 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of rectangle } ABED &= 6 \times 22.5 \\ &= 135 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of rectangle } ADFE &= 8 \times 22.5 \\ &= 180 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of rectangle } BEFC &= 10 \times 22.5 \\ &= 225 \text{ cm}^2\end{aligned}$$

Hence,

Surface area of the prism = Sum of the area of all five sides

$$= 24 + 24 + 135 + 180 + 225$$

$$= 588 \text{ cm}^2$$

Total: 11 marks

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7. The table below shows the speeds, to the nearest kmh^{-1} , of 90 vehicles that pass a checkpoint.

Speed (in kmh^{-1})	Frequency	Cumulative Frequency
0 – 19	5	5
20 – 39	11	16
40 – 59	26	$26 + 16 = 42$
60 – 79	37	$37 + 42 = 79$
80 – 99	9	$9 + 79 = 88$
100 – 119	2	$2 + 88 = 90$

- (a) For the class interval 20-39, as written in the table above, complete the following sentences.

(i) The upper class limit is 39 [1]

(ii) The class width is $(39.5 - 19.5) = 20$ [1]

(iii) Sixteen vehicles passed a checkpoint at no more than
..... 39.5 kmh^{-1} [1]

- (b) Complete the table shown above by inserting the missing values for the cumulative frequency column. [2]

Shown on the table above.

- (c) On the grid provided on page 21, using a scale of 2 cm to represent 20 kmh^{-1} on the x -axis, and 2 cm to represent 10 vehicles on the y -axis, draw the cumulative frequency curve to represent the information in the table. [4]

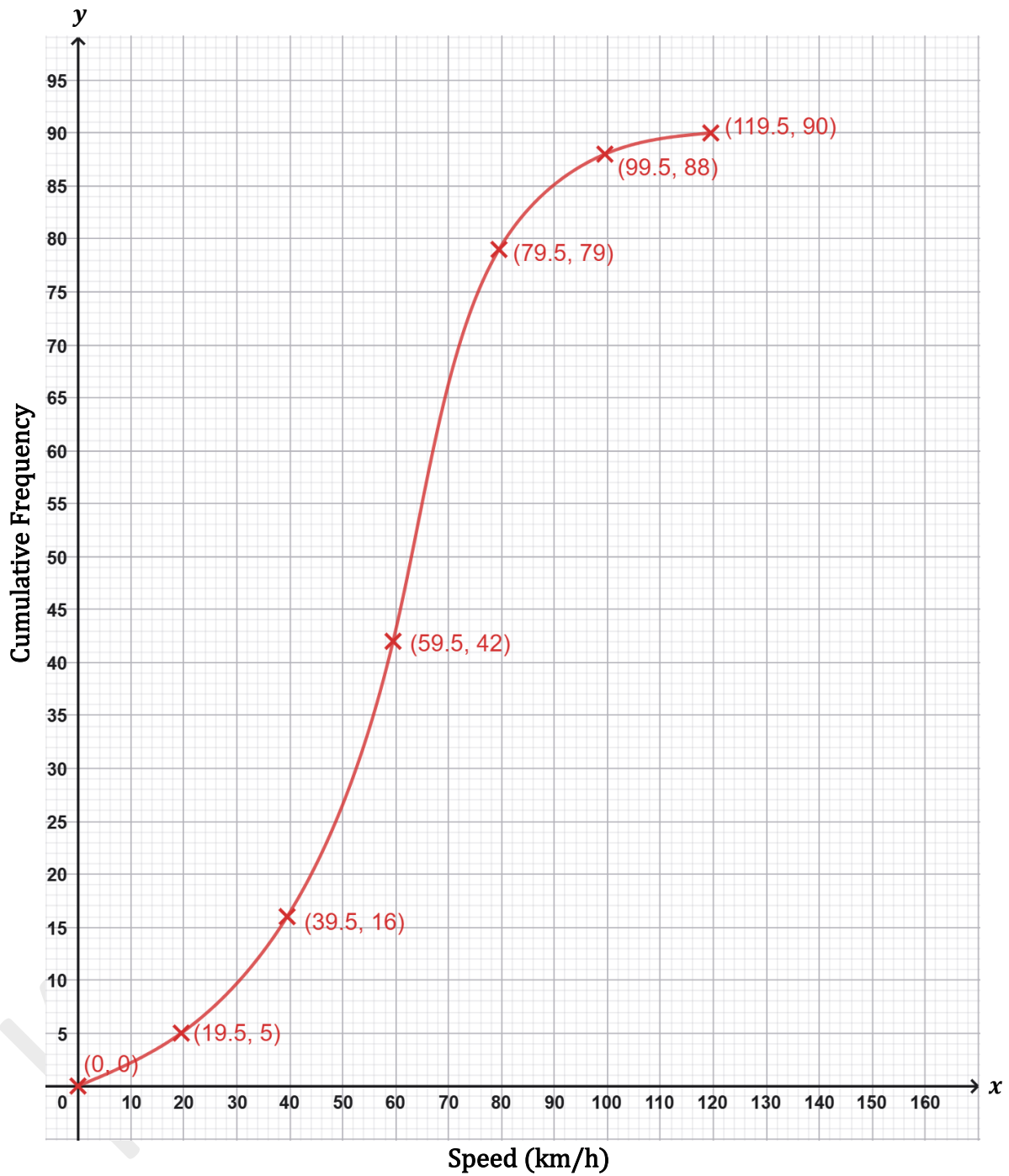
Speed (in kmh^{-1})	Frequency	Cumulative Frequency	Points to plot (U. C. B. , CF)
0 – 19	5	5	(19.5, 5)
20 – 39	11	16	(39.5, 16)
40 – 59	26	42	(59.5, 42)
60 – 79	37	79	(79.5, 79)
80 – 99	9	88	(99.5, 88)
100 – 119	2	90	(119.5, 90)

The cumulative frequency curve is shown on the next page.

Scale Used:

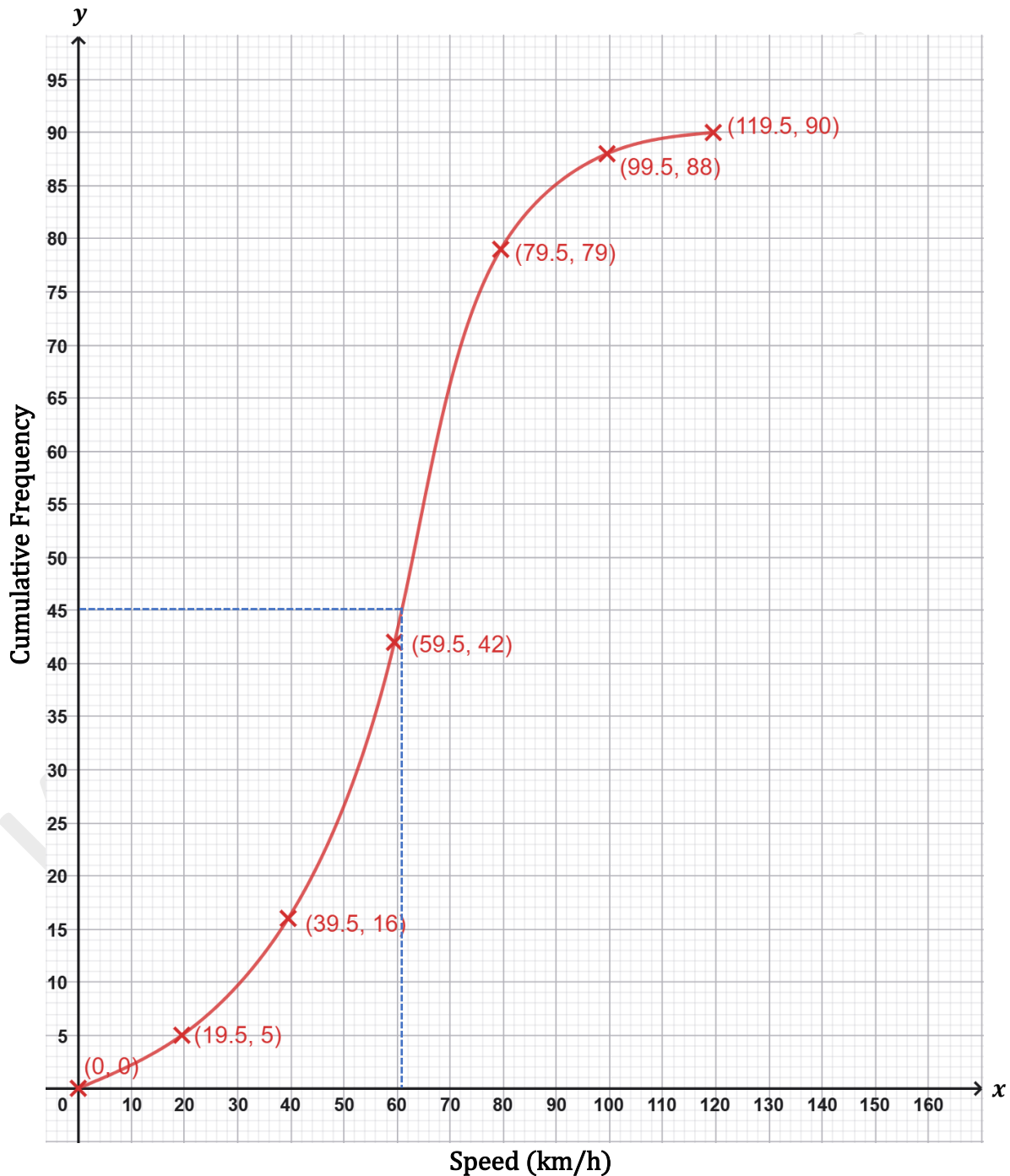
x -axis: 2cm represents 20 kmh^{-1} .

y -axis: 2cm represents 10 vehicles.



- (d) (i) On your graph, draw reference lines to estimate the speed at which no more than 50% of the vehicles drove as they passed the check point. [1]

50% of the 90 vehicles = 45 vehicles.



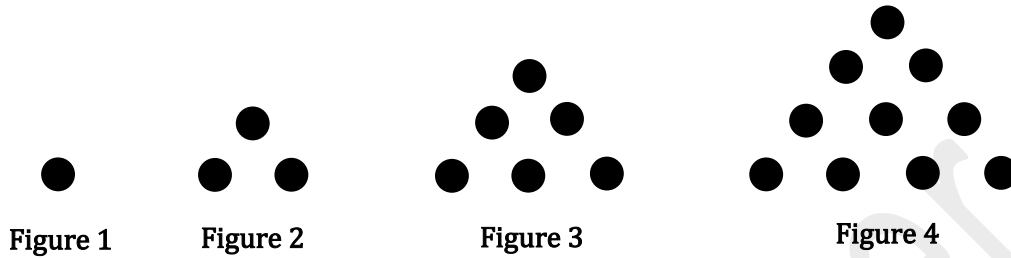
(ii) What is the estimated speed?

The estimated speed is 61 kmh^{-1} .

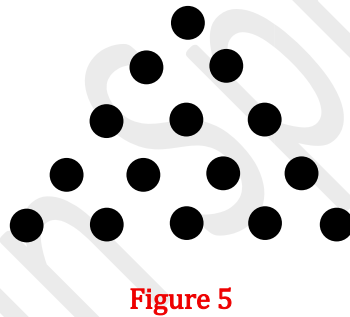
Total: 11 marks

Kerwin Springer

8. The first four figures in a sequence are shown below. Figure 1 is a single black dot, while each of the others consist of black dots arranged in an equilateral manner.



- (a) Draw Figure 5 of the sequence in the space below. [2]



- (b) How many dots would be in Figure 6? [1]

$$\begin{aligned} \text{The number of dots in Figure 6} &= 1 + 2 + 3 + 4 + 5 + 6 \\ &= 21 \text{ dots} \end{aligned}$$

The table below refers to the figures and the number of dots in each figure. Study the patterns shown.

Figure, n	Number of Dots, d , in terms of n	Number of Dots Used, d
1	$\frac{1}{2} \times 1 \times (1 + 1)$	1
2	$\frac{1}{2} \times 2 \times (2 + 1)$	3
3	$\frac{1}{2} \times 3 \times (3 + 1)$	6
$\vdots$		
11	$\frac{1}{2} \times 11 \times (11 + 1)$	66
$\vdots$		
n	$\frac{1}{2} \times n \times (n + 1)$	$\frac{1}{2} n(n + 1)$

(c) Complete the row which corresponds to Figure 11 in the table above. [2]

Shown on the table above.

(d) Determine which figure in the sequence has 210 dots. [2]

$$\frac{1}{2} n(n + 1) = 210$$

$$n(n + 1) = 420$$

$$n(n + 1) = 20(20 + 1)$$

$$\therefore n = 20$$

So, Figure 20 has 210 dots.

- (e) Write a simplified algebraic expression for the number of dots, d , in the Figure n . [1]

$$\begin{aligned} \text{In terms of } n, \text{ the number of dots, } d &= \frac{1}{2} \times n \times (n + 1) \\ &= \frac{1}{2} n(n + 1) \end{aligned}$$

(Shown on the table above)

- (f) Show that there is no diagram that has exactly 1000 dots. [2]

$$\text{Let } \frac{1}{2} n(n + 1) = 1000$$

$$n(n + 1) = 2000$$

There are no two consecutive integers, n and $n + 1$ being consecutive integers, whose product is exactly 2000.

Therefore, no diagram has 1000 dots.

Total: 10 marks

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The velocity-time graph below shows the motion of a cyclist over a period of 40 seconds.



- (i) Calculate the gradient of

(a) OA

[1]

Points are: $O(0,0)$ and $A(25,10)$

$$\text{Gradient of } OA = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10-0}{25-0}$$

$$= \frac{10}{25}$$

$$= \frac{2}{5}$$

(b) AB

[1]

Points are: $A(25, 10)$ and $B(40, 10)$

$$\text{Gradient of } AB = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10-10}{40-25}$$

$$= \frac{0}{15}$$

$$= 0$$

(ii) Complete the following statements.

The cyclist started from rest, where his velocity was 0 ms^{-1} , and steadily increased his velocity by $\frac{2}{5}$ ms^{-1} each second during the first 25 seconds.

During the next 15 seconds, his velocity remained constant, that is, his acceleration was 0 ms^{-1} . [3]

- (iii) Determine the average speed of the cyclist over the 40-second period.

[3]

Total distance = Area under the graph

$$= \frac{1}{2}[(40 - 25) + (40 - 0)] \times 10$$

$$= \frac{1}{2}(15 + 40) \times 10$$

$$= 275 \text{ m}$$

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}}$$

$$= \frac{275}{40}$$

$$= 6.875 \text{ ms}^{-1}$$

- (b) Consider the following pair of simultaneous equations:

$$x^2 + 2xy = 5$$

$$x + y = 3$$

- (i) WITHOUT solving, show that (1,2) is a solution for the pair of simultaneous equations.

[2]

Substituting (1, 2) into Equation 1 gives:

$$(1)^2 + 2(1)(2) = 1 + 4$$

$$= 5$$

Therefore, (1, 2) is a solution for Equation 1.

Substituting (1, 2) into Equation 2 gives:

$$1 + 2 = 3$$

Therefore, (1, 2) is a solution for Equation 2.

Hence, (1, 2) is a solution for the pair of simultaneous equations.

- (ii) Solve the pair of simultaneous equations above to determine the **other** solution. [5]

$$x^2 + 2xy = 5 \quad \dots \text{Equation 1}$$

$$x + y = 3 \quad \dots \text{Equation 2}$$

Rearranging Equation 2 gives:

$$y = 3 - x \quad \dots \text{Equation 3}$$

Substituting Equation 3 into Equation 1 gives:

$$x^2 + 2x(3 - x) = 5$$

$$x^2 + 6x - 2x^2 = 5$$

$$-x^2 + 6x = 5$$

$$x^2 - 6x + 5 = 0$$

$$x^2 - 5x - x + 5 = 0$$

$$x(x - 5) - 1(x - 5) = 0$$

$$(x - 1)(x - 5) = 0$$

$$\begin{array}{lll} \text{Either} & x - 1 = 0 & \text{or} & x - 5 = 0 \\ & x = 1 & & x = 5 \end{array}$$

We already know that $(1, 2)$ is one solution.

When $x = 5$,

$$y = 3 - 5$$

$$y = -2$$

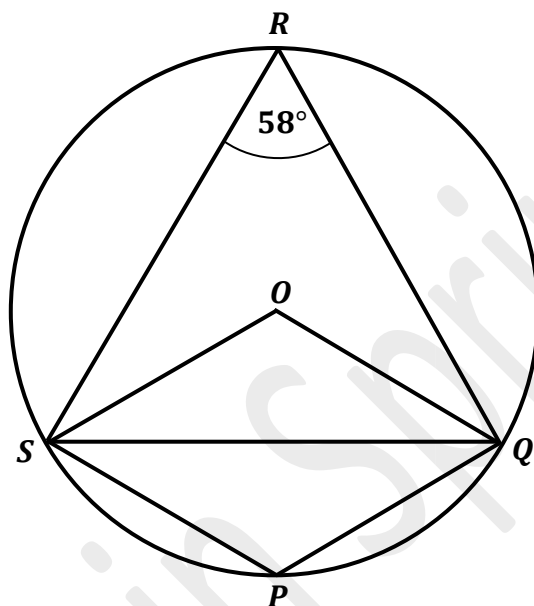
$\therefore$ The other solution is $(5, -2)$.

Total: 15 marks

MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) P, Q, R and S are four points on the circumference of the circle shown below.

Angle $QRS = 58^\circ$



Using the geometrical properties of a circle to give reasons for each step of your answer, determine the measure of

(i) $\angle SPQ$

[2]

The opposite angles of a cyclic quadrilateral are supplementary.

$$\angle SPQ = 180^\circ - 58^\circ$$

$$= 122^\circ$$

(ii) $\angle OQS$

Since OS and OQ are both radii, triangle OSQ is isosceles.

So,

$$\angle OSQ = \angle OQS$$

Now:

$$\begin{aligned}\angle SOQ &= 2 \times 56^\circ \\ &= 116^\circ\end{aligned}$$

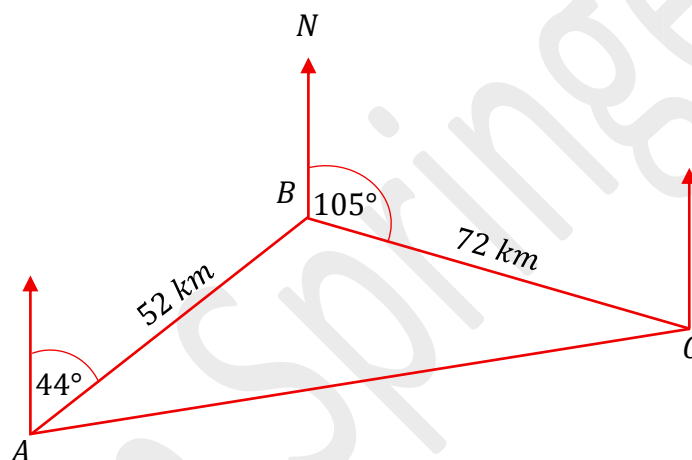
Angles in triangle OSQ add up to 180° :

$$\begin{aligned}\angle OQS &= \frac{180^\circ - 116^\circ}{2} \\ &= \frac{64^\circ}{2} \\ &= 32^\circ\end{aligned}$$

$$\therefore \angle OQS = 32^\circ$$

(b) A ship leaves Port A and sails 52 km on a bearing of 044° to Port B. The ship then changes course to sail to Port C, 72 km away, on a bearing of 105° .

- (i) On the diagram below, **not drawn to scale**, label the known distances travelled and the known angles. [2]



- (ii) Determine the measure of $\angle ABC$. [2]

Alternate angles are equal.

$$\angle ABS = 44^\circ$$

Two angles which make a straight line are supplementary.

$$\angle CBS = 180^\circ - 105^\circ$$

$$= 75^\circ$$

$$\begin{aligned}\therefore \angle ABC &= 44^\circ + 75^\circ \\ &= 119^\circ\end{aligned}$$

(iii) Calculate, to the nearest km, the distance AC .

[3]

By the cosine rule:

$$\begin{aligned}AC^2 &= (52)^2 + (72)^2 - 2(52)(72) \cos 119^\circ \\ &= 2704 + 5184 - (-3630.25) \\ &= 11518.25\end{aligned}$$

$$\begin{aligned}AC &= \sqrt{11518.25} \\ &= 107 \text{ km} \quad (\text{to the nearest km})\end{aligned}$$

(iv) Show that the bearing of A from C , to the nearest degree, is 260° . [3]

Co-interior angles are supplementary.

$$\angle BCN = 75^\circ$$

By the sine rule:

$$\begin{aligned}\frac{52}{\sin \hat{A}CB} &= \frac{107.3}{\sin 119^\circ} \\ \sin \hat{A}CB &= \frac{52 \sin 119^\circ}{107.3}\end{aligned}$$

$$\sin \hat{A}CB = 0.434$$

$$\begin{aligned}\hat{A}CB &= \sin^{-1}(0.434) \\ &= 25.07^\circ\end{aligned}$$

The bearing of A from $C = 360^\circ - (75^\circ + 25.07^\circ)$

$= 260^\circ$ (to the nearest degree)

Total: 15 marks

Kerwin Springer

VECTORS AND MATRICES

11. (a) Matrices A and B are such that

$$A = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \text{ and } B = \begin{pmatrix} 4 & 0 \\ 3 & -1 \end{pmatrix}.$$

(i) Show by multiplying A and B , that $AB \neq BA$. [2]

$$\begin{aligned} AB &= \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 3 & -1 \end{pmatrix} \\ &= \begin{pmatrix} (3 \times 4) + (2 \times 3) & (3 \times 0) + (2 \times -1) \\ (5 \times 4) + (4 \times 3) & (5 \times 0) + (4 \times -1) \end{pmatrix} \\ &= \begin{pmatrix} 12 + 6 & 0 - 2 \\ 20 + 12 & 0 - 4 \end{pmatrix} \\ &= \begin{pmatrix} 18 & -2 \\ 32 & -4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} BA &= \begin{pmatrix} 4 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \\ &= \begin{pmatrix} (4 \times 3) + (0 \times 5) & (4 \times 2) + (0 \times 4) \\ (3 \times 3) + (-1 \times 5) & (3 \times 2) + (-1 \times 4) \end{pmatrix} \\ &= \begin{pmatrix} 12 + 0 & 8 + 0 \\ 9 - 5 & 6 - 4 \end{pmatrix} \\ &= \begin{pmatrix} 12 & 8 \\ 4 & 2 \end{pmatrix} \end{aligned}$$

Since $\begin{pmatrix} 18 & -2 \\ 32 & -4 \end{pmatrix} \neq \begin{pmatrix} 12 & 8 \\ 4 & 2 \end{pmatrix}$, then $AB \neq BA$.

(ii) Find A^{-1} , the inverse of A .

$$\begin{aligned}\det A &= ad - bc \\ &= (3)(4) - (2)(5) \\ &= 12 - 10 \\ &= 2\end{aligned}$$

$$\begin{aligned}\text{adj } A &= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}A^{-1} &= \frac{1}{2} \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix} \\ &= \begin{pmatrix} \frac{4}{2} & \frac{-2}{2} \\ \frac{-5}{2} & \frac{3}{2} \end{pmatrix} \\ &= \begin{pmatrix} 2 & -1 \\ -\frac{5}{2} & \frac{3}{2} \end{pmatrix}\end{aligned}$$

- (iii) Write down the 2×2 matrix representing the matrix product

$$AA^{-1}.$$

[1]

$$AA^{-1} = I$$

$$\therefore I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- (b) (i) Write the following pair of simultaneous equations as a matrix equation.

$$3x + 2y = 1$$

$$5x + 4y = 5$$

[1]

$$3x + 2y = 1$$

$$5x + 4y = 5$$

In matrix form,

$$\begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

- (ii) Write the solution of your matrix equation in (b)(i) as a product of two matrices. [2]

$$\begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

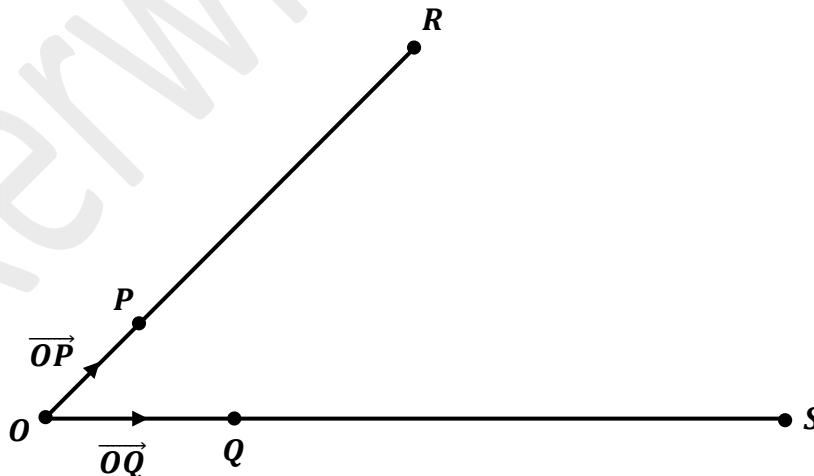
$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -\frac{5}{2} & \frac{3}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

- (c) The position vectors of the points P and Q relative to an origin, O , are

$$\overrightarrow{OP} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \text{ and } \overrightarrow{OQ} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} \text{ respectively.}$$

The diagram below shows that $PR = 3 OP$ and $QS = 3 OQ$.



(i) Express in the form $\begin{pmatrix} x \\ y \end{pmatrix}$, vector

• $\overrightarrow{OS}$

[1]

$$\overrightarrow{OQ} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

$$\begin{aligned} \overrightarrow{QS} &= 3 \begin{pmatrix} 5 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 15 \\ 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{OS} &= \overrightarrow{OQ} + \overrightarrow{QS} \\ &= \begin{pmatrix} 5 \\ 0 \end{pmatrix} + \begin{pmatrix} 15 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 20 \\ 0 \end{pmatrix} \text{ which is of the form } \begin{pmatrix} x \\ y \end{pmatrix}, \text{ where } x = 20 \text{ and } y = 0 \end{aligned}$$

• $\overrightarrow{PQ}$

[2]

$$\begin{aligned} \overrightarrow{PQ} &= \overrightarrow{PO} + \overrightarrow{OQ} \\ &= -\begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} 5 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ which is of the form } \begin{pmatrix} x \\ y \end{pmatrix}, \text{ where } x = 1 \text{ and } y = -3 \end{aligned}$$

- $\overrightarrow{RS}$

$$\overrightarrow{PR} = 3\overrightarrow{OP}$$

$$= 3 \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 12 \\ 9 \end{pmatrix}$$

$$\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{PR}$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} 12 \\ 9 \end{pmatrix}$$

$$= \begin{pmatrix} 16 \\ 12 \end{pmatrix}$$

$$\overrightarrow{RS} = \overrightarrow{RO} + \overrightarrow{OS}$$

$$\overrightarrow{RS} = -\begin{pmatrix} 16 \\ 12 \end{pmatrix} + \begin{pmatrix} 20 \\ 0 \end{pmatrix}$$

$$\overrightarrow{RS} = \begin{pmatrix} 4 \\ -12 \end{pmatrix}$$

which is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$, where $x = 4$ and $y = -12$

- (ii) State TWO geometrical relationships between PQ and RS .

$$\overrightarrow{PQ} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\overrightarrow{RS} = \begin{pmatrix} 4 \\ -12 \end{pmatrix}$$

$$= 4 \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$= 4 \overrightarrow{PQ}$$

Hence, $\overrightarrow{RS}$ is a scalar multiple of $\overrightarrow{PQ}$.

Therefore, $\overrightarrow{RS}$ and $\overrightarrow{PQ}$ are parallel.

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.